

Frank Llonch Valsells

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EDUCATION

- **Universidad Alfonso X el Sabio** Madrid, Spain
Bachelor of Mathematical Engineering · Final Degree Project with Ocean Ecostructures 2022 – 2026
- **Colegio Montserrat** Barcelona, Spain
International Baccalaureate (Honours) 2019 – 2021

EXPERIENCE

- **Sirt Consulting** Barcelona, Spain
Data Engineer Intern Nov 2025 – Mar 2026
 - **Product:** Rebuilt the software behind a long-running educational personality test: the assessment was a proven seller, the system running it was antiquated, so we modernised it end to end.
 - **Security:** Helped stand up a SOC (security operations centre) offering aimed at small and medium businesses.
- **DeNexus** Madrid, Spain
Cyber Data Analyst Intern Nov 2024 – Feb 2025
 - **Description:** Built a system that ingests cyber-incident disclosures from SEC filings and identifies and classifies the incidents reported inside them with deep learning: transformer embeddings and tokenisation (BERT) over long filing text, packaged with Docker.

PROJECTS

- **Melanomator · Melanoma Classification (CNNs):** Binary benign/malignant skin-lesion classifier in PyTorch: a custom CNN benchmarked against a ResNet18 transfer-learning baseline, with Grad-CAM heatmaps so every prediction is explainable. 92% accuracy and 87% recall on 9.6k held-out lesions; live at melanomapp.streamlit.app.
- **Cancer Mortality Prediction (Deep Learning):** A Keras/TensorFlow neural network predicting patient mortality from clinical biomarkers, tuned by a 90-trial Keras Tuner search over architecture, regularisation and learning rate.
- **WW2 RAG:** A fully offline retrieval-augmented chatbot: chunking and BGE sentence embeddings into an Elasticsearch vector store, cosine retrieval, and a local LLM served by Ollama behind a Streamlit UI. Runs 100% locally via Docker.
- **BBVA vs Santander:** Daily share-price forecasting for two Spanish banks: LSTM and GRU networks trained per bank in PyTorch, with fitted scalers and the best checkpoints served behind a Streamlit app.
- **Linear Optimisation:** Primal and dual linear programmes in Pyomo maximising agricultural profit, with shadow-price analysis to find the binding constraints.
- **Media Mix Model:** Ridge-regression attribution across four marketing channels, with ARIMA forecasting of web traffic, feeding a simulator that reallocates monthly budget to lift ROI.
- **Dimensional Warehouse & CLTV:** A PostgreSQL star schema built and loaded with integrity checks, then churn labelling, logistic regression with threshold tuning, and a five-year customer lifetime value per client, visualised in Plotly.
- **Purchase Propensity:** A Random Forest classifier that flags which customers are likely to buy a second vehicle.
- **PCA & Clustering:** PCA and K-means segmentation turning raw customer data into distinct, actionable profiles; the same pipeline applied to a thyroid clinical dataset.
- **De la biodiversidad al balance:** Final degree project with Ocean Ecostructures: a mathematical-financial model that turns reef biodiversity indicators into monetisable economic value, across three real ports with Monte Carlo uncertainty and an interactive dashboard.
- **Table Parsing Agent (A La Maison):** An agent on pydantic-ai that reads catalogues straight from images and returns schema-validated records, driven from a Google Sheet and deployed as a Cloud Function.
- **Other: Ticketator** (n8n and Azure Document Intelligence turning emailed receipts into Google Sheets rows), **Neomodernismo** (real-time shape recognition in OpenCV, Canny edges and contours, paired with generative geometry) and a **Django + TMDb streaming app** with auth and REST API. All at github.com/frankllonch.

SKILLS

- **Deep Learning:** CNNs and transfer learning (ResNet), Grad-CAM explainability, transformers and BERT embeddings, tokenisation, RAG and vector retrieval (Elasticsearch), LSTM and GRU sequence models
- **Machine Learning:** scikit-learn, Ridge and logistic regression, Random Forest, K-means and PCA, ARIMA and SARIMA, topic modelling, hyperparameter search (Keras Tuner)
- **Optimisation:** Linear and integer programming in Pyomo, duality and shadow-price sensitivity analysis
- **Frameworks & Data:** PyTorch, TensorFlow, torchvision, OpenCV, pandas, Jupyter, PostgreSQL, MongoDB, Docker, FastAPI, Streamlit, Databricks, ETL and dimensional modelling, Power BI, n8n
- **Programming:** Python, SQL, JavaScript, TypeScript, HTML/CSS

SOFT SKILLS

- **How I work:** Hard worker who takes a project end to end, from messy data to something a client can use. Creative problem solver and a people person, used to explaining technical work to non-technical audiences.

PERSONAL

- **Before engineering:** Studied filmmaking at ESCAC in Barcelona (still shoot and edit), draw whenever I can, and worked a season in Menorca as a PADI Dive Master.
- **Languages:** Spanish and Catalan (native), English (fluent, IGCSE), Portuguese (B1).